SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

Q.23 Find the arithmetic mean for the following data

Marks obtained	5	8	10	15	17
No. of Students	3	4	3	7	2

Q.24 Calculate the cloth cover or fabric cover of a fabric if wrap count is 20’s Ne, Weft is 20’s Ne EPI= 80 and PPI = 60

Q.25 Define periodic variation in textile. How will you classify it. Explain how this classification helps in to detect the machine which cause the variation.

No. of Printed Pages : 4

Roll No. ....

3rd Sem / Textile Design

Subject : Textile Mathematics

Time : 3 Hrs. M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

Q.1 Value of 0!

a) 0 b) 1

c) 2 d) 3

Q.2 Value of $\log_2 2$ is

a) 1 b) 2

c) 3 d) 4

Q.3 Simplest ratio of 10 :20

a) 1:2 b) 2:1

c) 4:5 d) 5:4

Q.4 The perimeter of rectangle is

a) Length x Breadth b) 2 (Length + breadth)

c) Length + Breadth d) none of these

Q.5 Value of ${}^{1}P_{1}$

a) 0 b) 1

c) 2 d) 3

Q.6 The Mode of the series

7,5,2,5,3,7,7, is

a) 2 b) 5

c) 3 d) 7

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

Q.7 Find the median of the series 2,3,7,5,8.

Q.8 Write the formula of the area of triangle.

Q.9 A mixing of cotton is made of two type of cotton A and B. The ratio of A and B is 2:4 If total mixing is 130 kg. Then find weight of each type of cotton.

Q.10 Find the area of rectangle whose length is 10 m and breadth is 5 m.

Q.11 Value of ${}^{n}C_{r}$ is.

Q.12 If a fabric is having EPI 35 and warp count is 26's Ne. Calculate the warp cover factor of this fabric.

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

Q.13 Find fourth proportion of 9,16,25.

Q.14 In how many ways the letter in a word TWO can be arranged.

Q.15 Define random variation in brief.

Q.16 Discuss the use of control chart in brief.

Q.17 Find the fencing of a square park of length 80 m at the rate of Rs 15 per meter.

Q.18 Find the value of x if log(x+2) - log (5x+3) = log7.

Q.19 Find the diagonal of the rectangle whose length is 25 cm and breadth is 15 cm.

Q.20 Find the area of triangle whoe height is 12 cm and base is 13 cm.

Q.21 If (n+1)! = 5(n-1)!, then find the value of n.

Q.22 If the mean of x+1, x+3, x+5, x+7, x+9 is 10 then find the mean of first three numbers.